

MID-CAROLINA, NC, USA

WMO: 723156

Lat: 35.646N

Lon: 80.520W

Elev: 236

StdP: 98.53

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 63812

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.4	-5.4	-19.4	0.7	-1.1	-16.4	0.9	0.2	8.7	10.5	7.7	9.9	1.1	10	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	34.0	22.9	32.7	22.9	32.0	22.7	25.3	30.2	24.8	29.6	24.2	28.8	2.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.0	19.5	26.7	23.4	18.7	26.2	22.8	18.1	25.8	78.6	30.3	76.3	29.5	74.2	28.4	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.1	5.3	-11.8	35.4	2.9	1.8	-14.0	36.7	-15.7	37.7	-17.3	38.7	-19.5	40.0		
(5)			-12.3	26.7	2.7	0.9	-14.2	27.3	-15.9	27.9	-17.4	28.4	-19.4	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.4	4.6	6.3	10.3	15.4	19.6	23.8	25.7	25.1	21.9	15.8	9.6	6.7		
	DBStd	8.57	5.74	5.26	5.16	4.46	3.83	2.65	2.14	2.54	3.14	4.41	4.68	4.82		
	HDD10.0	579	186	127	62	8	0	0	0	0	0	6	63	128		
	HDD18.3	1901	427	338	254	108	34	1	0	0	8	104	265	362		
	CDD10.0	2565	18	24	70	169	297	414	485	469	358	185	51	25		
	CDD18.3	844	1	1	4	20	73	166	227	211	116	24	2	1		
	CDH23.3	7500	1	4	33	219	648	1496	2091	1856	916	213	18	3		
	CDH26.7	2851	0	0	2	36	180	582	911	777	314	47	2	0		
(14) Wind	WSAvg	1.8	2.1	2.3	2.5	2.5	2.0	1.6	1.5	1.3	1.5	1.6	1.7	1.7		
(15) Precipitation	PrecAvg	1053	86	90	104	82	86	100	104	85	88	84	71	80		
	PrecMax	1525	184	202	214	271	183	253	256	200	267	356	135	164		
	PrecMin	672	22	14	25	9	11	2	25	15	0	0	10	15		
	PrecStd	170	42	42	47	48	43	59	54	45	58	72	35	42		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.0	23.8	26.7	29.1	32.2	34.8	36.0	37.1	33.8	31.1	26.0	22.6		
		MCWB	15.4	16.2	15.2	17.9	21.2	22.8	22.6	23.1	20.9	21.9	17.0	17.1		
	2%	DB	18.7	21.0	23.8	27.6	30.2	32.8	33.9	33.9	32.2	27.7	22.5	19.4		
		MCWB	14.1	14.0	14.5	17.0	20.4	22.5	23.5	22.8	21.1	18.7	15.6	16.9		
	5%	DB	17.1	18.0	22.1	26.2	28.8	32.0	32.7	32.4	30.1	25.9	21.0	17.5		
		MCWB	12.9	12.6	14.0	16.5	20.2	22.3	23.5	23.3	21.5	18.9	15.2	14.2		
	10%	DB	14.1	15.9	19.1	24.0	27.3	30.1	31.6	31.1	28.0	23.5	18.0	15.3		
		MCWB	9.7	10.7	12.7	15.9	19.4	21.9	23.2	23.1	20.8	18.0	13.4	11.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.5	18.1	18.0	20.8	23.0	25.6	26.2	26.0	24.6	23.1	21.1	19.4		
		MCDB	18.7	21.7	22.4	24.9	27.1	29.4	29.9	30.1	28.5	26.8	22.1	20.7		
	2%	WB	15.6	16.4	16.8	19.3	22.2	24.5	25.4	25.3	23.9	22.2	18.1	17.4		
		MCDB	17.9	18.7	20.5	23.7	26.9	29.3	30.5	30.0	27.9	25.7	20.1	19.1		
	5%	WB	13.6	13.9	15.7	18.1	21.4	23.9	24.8	24.7	23.1	20.7	16.4	15.5		
		MCDB	16.2	16.7	19.6	22.7	26.0	28.7	29.9	29.3	26.7	23.7	19.1	17.0		
	10%	WB	10.8	11.3	14.3	17.2	20.6	23.2	24.2	24.2	22.4	19.3	14.6	12.6		
		MCDB	13.6	14.6	18.3	21.7	25.0	27.8	28.9	28.6	25.9	22.5	17.9	14.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	11.3	12.3	12.8	12.2	11.4	10.8	10.5	11.1	12.2	13.0	11.1		
		MCDBR	13.4	14.6	14.5	14.9	14.1	13.2	12.3	12.5	13.7	14.0	15.6	13.0		
	5% WB	MCWBR	9.1	9.2	7.5	6.4	5.6	4.6	4.2	4.1	4.9	6.8	9.3	9.0		
		MCWBR	11.1	12.2	11.3	11.6	11.4	11.0	10.6	10.3	10.2	10.8	11.0	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.310	0.322	0.357	0.400	0.453	0.483	0.534	0.516	0.424	0.377	0.326	0.309		
	taud		2.498	2.461	2.399	2.277	2.165	2.119	2.002	2.031	2.283	2.390	2.494	2.535		
	Ebn at Noon		893	921	912	884	838	810	766	770	832	844	866	873		
	Edh at Noon		82	96	111	132	150	157	176	167	124	102	82	75		
(44) All-Sky Solar Radiation	RadAvg		2.58	3.23	4.31	5.39	5.99	6.43	6.00	5.47	4.60	3.77	2.86	2.21		
	RadStd		0.17	0.40	0.30	0.42	0.49	0.46	0.40	0.26	0.41	0.48	0.23	0.20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (72 neighbors)		+0.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+156	+104

Nomenclature: See separate page